

Answer to Reviewer 1

We would like to thank the Reviewer for their time and valuable suggestions. In what follows, we list the comments as numbered by the Reviewer and provide an answer to each. Together with the revised manuscript, we provide a version with tracked changes. The edited or added parts are highlighted in a blue font, while the deleted parts are crossed out. During the revision, some small changes in form or grammar have been implemented. For completeness, they are also included in the tracked version.

1. Clarification of the new insight

In the Introduction, we now explicitly mention the previous theoretical formulation by Johansen and state that it is not directly applicable to Mach–Zehnder interferometry. The last paragraph of Subsection 2.2 has been moved to the Discussion, where we explain in more detail why Johansen’s approach cannot be directly implemented for the path observables in a Mach–Zehnder interferometer and emphasize that the presented scheme overcomes this constraint.

2. Notation and normalization of the intensities

We thank the reviewer for pointing out this inconsistency. The notation in the Methods section has now been corrected and made consistent.

3. Resource comparison

A quantitative comparison with a previous neutron interferometer experiment in which a conventional weak measurement is reported (Ref. [21]) has been added in the second paragraph of the Discussion. The comparison highlights the reduction in experimental complexity and acquisition time. We also mention how the absence of spin analysis relaxes some crucial experimental limitations, e.g., beam cross-section, allowing for potential optimization of the setup to further reduce statistical uncertainty and/or measurement time.

Furthermore, we are particularly grateful for this comment, as it led us to identify a minor issue in the computation of the error-bars. Specifically, due to the difference in acquisition time between the measurements of relative path intensities $I_{+,j}$ and interferograms $I_+(\delta)$, a normalization had to be applied. However, for the relative path intensities, the statistical uncertainties have been calculated after the normalization, resulting in a slight overestimation of the statistical uncertainty. The mistake has now been fixed and the error-bars are slightly reduced.

4. Treatment of the contrast factor

We have revised the Methods section to provide a clearer physical justification of the correction model. The revised text now discusses the origin of the imperfect interference contrast, despite the nearly ideal beams produced at the first plate by Bragg diffraction.

5. Overstatement of the nonclassicality claim

The statement in Sec. 2.4 has been revised. It is now stated that the obtained measurement results do not reveal any nonclassical behavior with respect to the choice of initial states, post-selections and path observables.

Finally, we thank once more the Reviewer for the helpful comments and hope that we have now produced a more balanced and better account of our work. We hope that the revised manuscript is acceptable for publication in *npj Quantum Information*.

Best Regards,

Ismaele Vincent Masiello*, Hartmut Lemmel, Andreas Dvorak, Stephan Sponar, and Yuji Hasegawa

(*corresponding author)

Answer to Reviewer 2

We would like to thank the Reviewer for their time and valuable suggestions. In what follows, we list the comments as numbered in the report and provide an answer to each. Together with the revised manuscript, we provide a version with tracked changes. The edited or added parts are highlighted in a blue font, while the deleted parts are crossed out. During the revision, some small changes in form or grammar have been implemented. For completeness, they are also included in the tracked version.

0. General comment: elaboration of the novelties of the work

A quantitative comparison with a previous neutron interferometer experiment using a conventional weak measurement has been added in the Discussion. This highlights the reduction in experimental complexity and acquisition time. We also mention how the absence of spin analysis relaxes some crucial experimental limitations, e.g., beam cross-section, allowing for potential optimization of the setup to further reduce statistical uncertainty and/or measurement time.

1. Caption of Fig. 5a

The text in Sec. 2.4 has been adjusted and it is now consistent with the figure.

2. The balanced imaginary-part panels in Fig. 5b

An inset has now been added to the figure, better displaying the region of interest and making it comparable with the range of the unbalanced-configuration plot. We deeply appreciate the suggestion, nevertheless, we would like to express a preference for the original version of the plot. We leave the final decision to the Reviewer and the Editor.

Furthermore, a minor issue in the computation of the error-bars has been identified during the revision. Specifically, due to the difference in acquisition time between the measurements of relative path intensities $I_{+,j}$ and interferograms $I_+(\delta)$, a normalization had to be applied. However, for the relative path intensities, the statistical uncertainties have been calculated after the normalization, resulting in a slight overestimation of the error-bars. The mistake has now been fixed and the error-bars are slightly reduced.

Finally, we thank once more the Reviewer for the helpful comments and hope that we have now produced a more balanced and better account of our work. We hope that the revised manuscript is acceptable for publication in *npj Quantum Information*.

Best Regards,

Ismaele Vincent Masiello*, Hartmut Lemmel, Andreas Dvorak, Stephan Sponar, and Yuji Hasegawa

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